History-dependent readout tree

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Model

Let the null have fresh independent ideal bits $Z_1, \dots, Z_4$, each with success probability $s = 131/320$. Let Y_i be the reported bit, and let $w = Y_1 \cdots Y_{i-1}$ denote a prior reported prefix. At *every* position and every positive-probability reported prefix, assume

$$\Pr(Y_i = 1 \mid Z_i = 0, w) \le \varepsilon, \qquad \Pr(Y_i = 0 \mid Z_i = 1, w) \le \varepsilon,$$

where the probabilities are those of the same null process and the ideal bit is fresh even after conditioning on w: $\Pr(Z_i = 1 \mid w) = s$. Independence of the ideal bit from reported history is a substantive condition. Nothing here bounds a quantum instrument's postmeasurement state or guarantees that the same caps hold in different device time blocks or experimental paths.

At one node, the next reported-one rate $q(w) = \Pr(Y_i = 1 \mid w)$ obeys the sharp interval

$$a_\varepsilon = s(1-\varepsilon) \ \le q(w) \le \ b_\varepsilon = s + (1-s)\varepsilon. \tag{1}$$

The interval is also sufficient, because any $q \ge s$ is realized by false-positive probability $(q - s)/(1 - s)$ and zero false negatives, while any $q \le s$ is realized by false-negative probability $(s-q)/s$ and zero false positives. Choosing such a channel separately after each observed prefix realizes the entire corresponding tree law by induction. These are classical, causal reporting rules; they need not be physically available as a particular quantum readout.

The stipulated alternative is fixed throughout the analysis. Write

$$e_0 = \frac{1}{100}, \quad m = s + (1-2s)e_0 = \frac{6579}{16000}, \quad A = \frac{1}{4}\left[\frac{37}{100}\frac{15}{16}(1-2e_0)\right]^4.$$

For a length-four binary word y containing j ones, its probability is

$$Q(y) = m^j(1-m)^{4-j} - A(-1)^j. \tag{2}$$

All proper marginals of this alternative equal the corresponding product Bernoulli(m) marginals.

Proposition: exact whole-word compatibility threshold

Claim (proved for the specified finite model). The law Q in (2) can be reproduced exactly by a four-step causal reporting process with the independent ideal rate s and a common upper cap ε on both error directions at *every* reported history and position **if and only if**

$$\boxed{\varepsilon \ge \varepsilon_{\text{full}} = \frac{10396360128757}{92107698697800} = 0.1128717824431468\ldots.} \tag{3}$$

Proof. Marginalizing at least one future bit in (2) cancels the alternating term. At depths 0, 1, and 2, each conditional next-one rate is therefore m. At depth 3, for a prefix w with j ones,

$$Q(w) = m^j(1-m)^{3-j}, \quad q(w) = m + \frac{(-1)^j A}{m^j(1-m)^{3-j}}. \tag{4}$$

Every target prefix has positive mass. By (1), a necessary and sufficient global cap is the largest of the one-node requirements

$$\max\left\{0, \frac{q(w)-s}{1-s}, \frac{s-q(w)}{s}\right\}$$

across all 15 prefixes. Among the last-step prefixes, the four distinct one-rates and their required bounds are:

Number of prefix ones	Target next-one rate	Least directional cap
0	0.427540794200	0.030756900233
1	0.387769900872	0.052775814663
2	0.444721046342	0.059845157827
3	**0.363168114062**	**0.112871782443**

Earlier positions require only $(m-s)/(1-s) = 29/9450 \approx 0.003068783$. The largest of all requirements is therefore at $w = 111$. Its next-one probability is exactly

$$q(111) = \frac{81711338569043}{224995905216000} = 0.3631681140623368\ldots,$$

so $(s - q(111))/s$ reduces to (3). Necessity follows because a smaller cap violates (1) at this positive-mass prefix. At the threshold all nodes satisfy (1), and the explicit one-sided flipping construction above realizes every conditional rate in (4), proving sufficiency. □

The proposition is *not* a claim that the supplied alternative itself uses this large error rate. It instead quantifies how much freedom a distinct classical readout model would require to impersonate the same observed data.

What parity misses

Let O_i be the probability of odd parity after i reports under an allowed null, with $O_0 = 0$. At each step, choose reported-one rate b_ε following an even prefix and a_ε following an odd prefix. Since $s < 1/2$, this attains the sharp recurrence

$$O_i = b_\varepsilon + (1 - a_\varepsilon - b_\varepsilon)O_{i-1},$$

and hence

$$O_4^{\max}(\varepsilon) = b_\varepsilon \frac{1-[(1-2s)(1-\varepsilon)]^4}{1-(1-2s)(1-\varepsilon)}. \tag{5}$$

The alternative has $Q(\text{odd}) = 0.526209351956162\ldots$. Equation (5) first reaches that value at

$$\boxed{\varepsilon_{\text{parity}} \approx 0.044121657500400.} \tag{6}$$

At and above this smaller allowance, an allowed null can match the **single odd-parity probability**. Therefore parity-only population observations cannot distinguish the alternative from the null class. Nevertheless, throughout

$$0.044121657500400 \lesssim \varepsilon < 0.112871782443147,$$

the full law remains incompatible with the null because the 111 prefix still violates its conditional lower bound. This is an exact distinction between compressing the four-bit output to one parity bit and retaining the whole word. It does not prescribe a finite-shot optimal full-word test.

A predeclared bounded score detecting this surviving direction is

$$X_\varepsilon(W, Y) = \mathbf{1}\{W = 111\}(a_\varepsilon - Y).$$

Every admissible null satisfies $\mathbb{E}X_\varepsilon \le 0$. Under the target, $\mathbb{E}_Q X_\varepsilon = Q(111)[a_\varepsilon - q(111)]$, positive whenever $\varepsilon < \varepsilon_{\text{full}}$. For example this population excess is 0.00178935619623 at a **5%** directional cap. A small excess and a relatively rare prefix make the corresponding finite-sample efficiency a separate question.

Calibration uncertainty changes the statistical accounting

The four-step tree contains $2(1 + 2 + 4 + 8) = 30$ directional history cells. The familiar 6,343 figure assumes the cap is already valid, uses a fixed 5% hypothesis-test size, and omits the cost of checking reference preparations and readout under those conditions.

For a deliberately optimistic illustration, suppose each cell has n genuinely independent known-ideal reference trials, **zero** observed directional mistakes, stable behavior, and faithful transport to the device's gate path. A one-sided exact binomial upper bound at per-cell failure allocation $\gamma/30$ is

$$u_n = 1 - (\gamma/30)^{1/n}.$$

To place all 30 upper limits at or below 1% using the union bound with total calibration allowance $\gamma = 0.01$, the zero-error case requires at least

$$n = \left\lceil \frac{\log(0.01/30)}{\log(0.99)} \right\rceil = 797$$

observations **in each cell**, or 23,910 classified cell observations across the 30 cells. This is not 23,910 necessarily distinct four-use records; records can contribute observations at several positions, and rare histories need sufficient representation. Even this optimistic criterion is unlikely to succeed when actual error rates sit near the 1% cap, and it does not prove stationarity or instrument transport.

If a total false-rejection budget of 5% is split into 1% for possible calibration-coverage failure and 4% for a held-out fixed-horizon parity test, the latter's first 95%-power crossing moves from 6,343 to **6,758** independent four-use records, assuming the population alternative and all conditional test premises. At 6,757 records the recalculated test power is 0.949974789382334; at 6,758 it is 0.950018042462656. This is a mathematical design illustration, not a physically justified or globally optimal photon or preparation budget. Allocation of coverage and multiple testing must be registered before collecting the held-out data.

Most importantly, even perfect past calibration cannot establish arbitrary future-history bounds without a justified stability or exchangeability/conditional model. For a true error rate exactly at the 1% boundary, a valid upper confidence limit does not typically fall below 1%; a practical certification task needs a real performance margin and directly relevant calibration data.